

Elionel Vertus

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Electrical Engineering — LEAD Scholars + EXCEL COMPASS

Expected Spring 2027

Relevant Coursework: Digital Systems, Digital Electronics, Electronics I & II, Linear Circuits, Engineering Statistics, Intro to Power Systems, Embedded Systems

EXPERIENCE

Undergraduate Researcher

Jan. 2026 – Present

University of Central Florida

Orlando, FL

- Develop **STM32 firmware** for embedded data acquisition, interfacing **IMU, encoder, LiDAR, and depth-camera** sensors used in real-time state-estimation experiments.
- Integrate STM32 sensing hardware with a **Raspberry Pi** processing platform for sensor collection, preprocessing, logging, and **multi-sensor data acquisition**.
- Support evaluation of an **Attentive Neural Process** against a ConvLSTM baseline, reaching **0.0016 testing RMSE vs. 0.0071**, approximately **77.5% lower prediction error**.

Firmware Co-Lead

Fall 2026 – Present

Project SHRIKE — IEEE Computer Society at UCF

Orlando, FL

- Co-lead firmware development for an embedded platform using the **NVIDIA Jetson Orin Nano**, focusing on **Linux boot-up, BSP configuration, and carrier-board bring-up**.
- Coordinate firmware and PCB integration for **power sequencing, GPIO, peripheral communication, and hardware connectivity**.
- Develop and debug the embedded Linux environment for **device configuration, peripheral testing, and system-level hardware debugging**.

PROJECTS

Wind Farm Transmission System | *PowerWorld, Power Flow, Contingency Analysis*

- Designed and evaluated a **200 MW wind-farm transmission system**, analyzing approximately **50 transmission lines** and **57 N-1 contingency cases**.
- Eliminated system violations and optimized **4 transmission-line additions totaling \$8.759M**, while evaluating an alternative configuration with approximately **\$1M lower capital cost**.

FM Modulation & PLL Demodulation System | *XR2206, LM565, TL084*

- Built and tested an **FM modulation and PLL demodulation system** using an **XR2206 VCO, LM565 PLL, and TL084 op-amp**.
- Measured **frequency deviation, modulation index, and recovered waveforms** using an R&S RTM3004 oscilloscope and recovered the original signal through **PLL demodulation**.

AI-Assisted CNC Robotic Arm | *Project Co-Lead, IEEE at UCF*

Oct. 2026 – Present

- Co-lead development of a **5-axis CNC robotic arm**, overseeing mechanical design, electronics, embedded firmware, motion planning, AI integration, and system validation.
- Lead system integration planning for an **Arduino Uno R4** and **Raspberry Pi** platform combining stepper-motor control, camera-based position feedback, and AI-assisted path correction.
- Coordinate subsystem requirements and testing plans for **inverse kinematics, joint limit switches, emergency-stop functionality, and collision detection**.

TECHNICAL SKILLS

Languages: C, C++, Python, ARMv7 Assembly, MATLAB, R, JavaScript

Embedded & Hardware: STM32, Raspberry Pi, Bare-Metal Programming, Microcontrollers, Soldering, Breadboarding, Digital Circuits

Engineering Tools: Vivado, Multisim, SolidWorks, Autodesk Inventor, Oscilloscope, Function Generator, Digital Multimeter